

Characteristics for the Wave Equation

Unit Speed and Speed $\pm c$

Dr. Allan Struthers

Michigan Technological University

Why Characteristics for the Wave Equation?

- The wave equation is second order
- But information still propagates along curves
- Characteristics reveal:
 - finite propagation speed
 - left- and right-moving waves
 - domain of dependence
- Everything reduces to transport equations

The 1D Wave Equation (Unit Speed)

$$u_{tt} - u_{xx} = 0$$

- Models vibrations of strings, acoustics, EM waves
- Second order in both space and time
- Hidden first-order structure

Characteristics $x \pm ct = \text{constant}$ via Substitution.

Consider a curve $x = x(t)$ in spacetime and define

$$U(t) := u(x(t), t).$$

By the chain rule,

$$U'(t) = u_t + x'(t) u_x.$$

Differentiate once more:

$$U''(t) = u_{tt} + 2x'(t)u_{xt} + (x'(t))^2 u_{xx} + x''(t)u_x.$$

For the wave equation $u_{tt} - c^2 u_{xx} = 0$, choose

$$x'(t) = \pm c, \quad x''(t) = 0.$$

Then $\frac{d^2 U}{dt^2} = u_{tt} - c^2 u_{xx} = 0$.

Conclusion: u is linear along characteristics because

$$x \pm ct = \text{constant} \implies \partial_{tt}(u(x(t), t)) = 0.$$

Factorization

Observe:

$$u_{tt} - u_{xx} = (\partial_t - \partial_x)(\partial_t + \partial_x)u$$

Define

$$v = u_t + u_x, \quad w = u_t - u_x$$

Then

$$\begin{cases} v_t - v_x = 0 \\ w_t + w_x = 0 \end{cases}$$

Interpretation

- The wave equation splits into two transport equations
- Each travels along its own family of characteristics
- One moves right, one moves left

Key Insight

Wave motion = superposition of left- and right-moving signals.

Right-Moving Characteristics

For

$$v_t - v_x = 0$$

Characteristics satisfy

$$\frac{dx}{dt} = 1 \quad \Rightarrow \quad x - t = \text{constant}$$

- Right-moving waves
- Constant shape, rigid transport

Left-Moving Characteristics

For

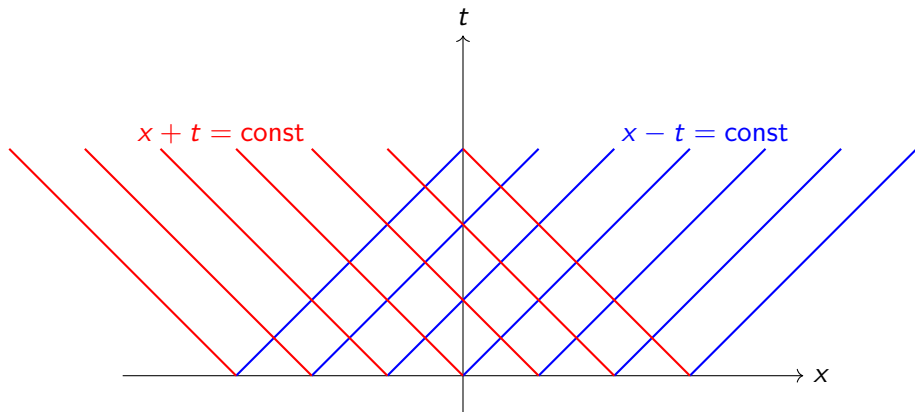
$$w_t + w_x = 0$$

Characteristics satisfy

$$\frac{dx}{dt} = -1 \quad \Rightarrow \quad x + t = \text{constant}$$

- Left-moving waves
- Independent of right-moving component

Spacetime Picture (Unit Speed)



Two independent families of characteristics.

General Solution

The general solution of

$$u_{tt} - u_{xx} = 0$$

is

$$u(x, t) = f(x - t) + g(x + t)$$

- f : right-moving wave
- g : left-moving wave
- No distortion, no steepening

Key Contrast with Burgers' Equation

- Linear equation
- Characteristics never intersect
- Superposition principle holds
- No shock formation

Geometry Matters

Crossing characteristics $\Rightarrow$ breakdown No crossing $\Rightarrow$ global smooth solutions

Wave Equation with Speed c

$$u_{tt} - c^2 u_{xx} = 0$$

- $c > 0$ is the wave speed
- Determines slope of characteristics
- Faster propagation for larger c

Factorization

$$u_{tt} - c^2 u_{xx} = (\partial_t - c\partial_x)(\partial_t + c\partial_x)u$$

Define

$$v = u_t + cu_x, \quad w = u_t - cu_x$$

Then

$$\begin{cases} v_t - cv_x = 0 \\ w_t + cw_x = 0 \end{cases}$$

Characteristic Speeds

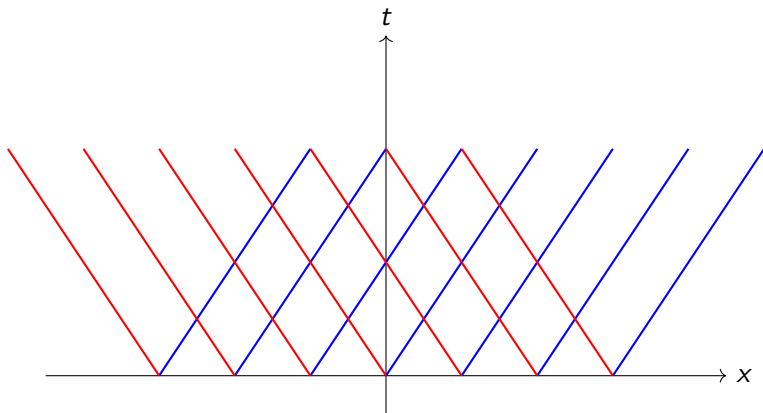
- Right-moving characteristics:

$$\frac{dx}{dt} = c \quad \Rightarrow \quad x - ct = \text{constant}$$

- Left-moving characteristics:

$$\frac{dx}{dt} = -c \quad \Rightarrow \quad x + ct = \text{constant}$$

Spacetime Picture (Speed $\pm c$)



Slope controlled by the wave speed c .

Domain of Dependence

- Value at (x, t) depends only on data in

$$[x - ct, x + ct]$$

- Finite propagation speed
- Sharp contrast with elliptic equations

Summary

- Wave equation hides two transport equations
- Two characteristic families: left and right
- Speed c controls characteristic slopes
- No characteristic crossing
- Geometry explains causality and propagation